AHDC calibration meeting

Recap : waveform classification, decoding, hit selection

Felix Touchte Codjo

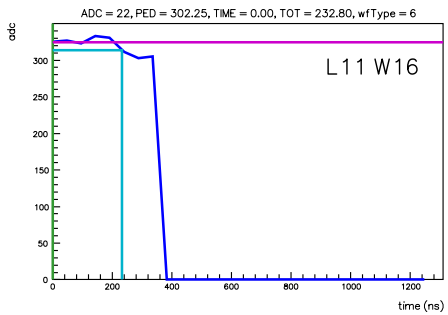
IJCLab

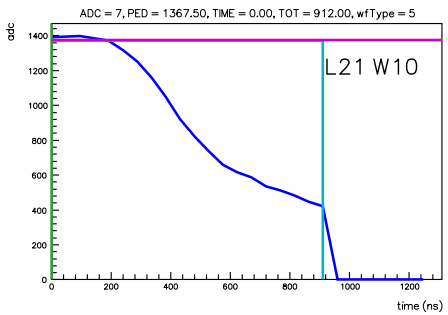
November 4, 2026

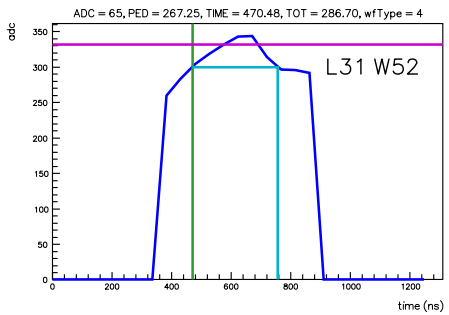
Waveform classification

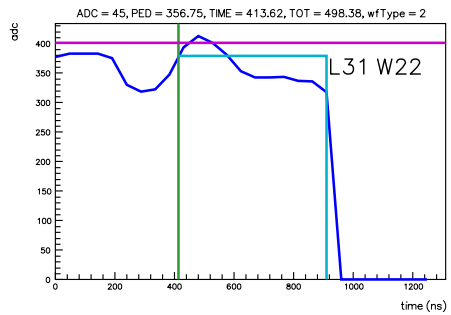
- The signals recorded by the AHDC have various shapes
- These different waveforms need specific treatments
- We gather them into seven categories

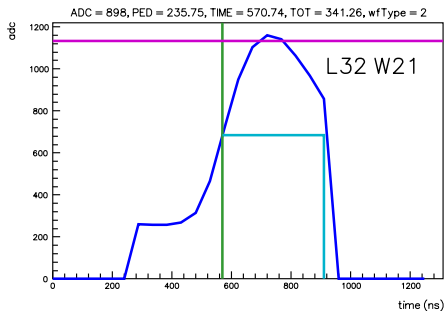
wfType 6 → too short
wfType 5 → decreasing
wfType 4 → small ToT
wfType 3 → pileup (ongoing work)
wfType 2 → missing trailing edge
wfType 1 → saturated
wfType 0 → normal

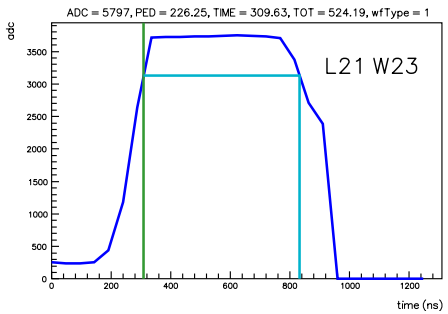


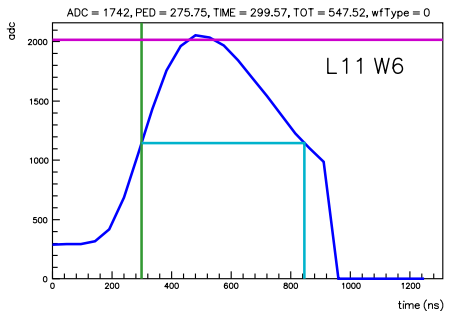








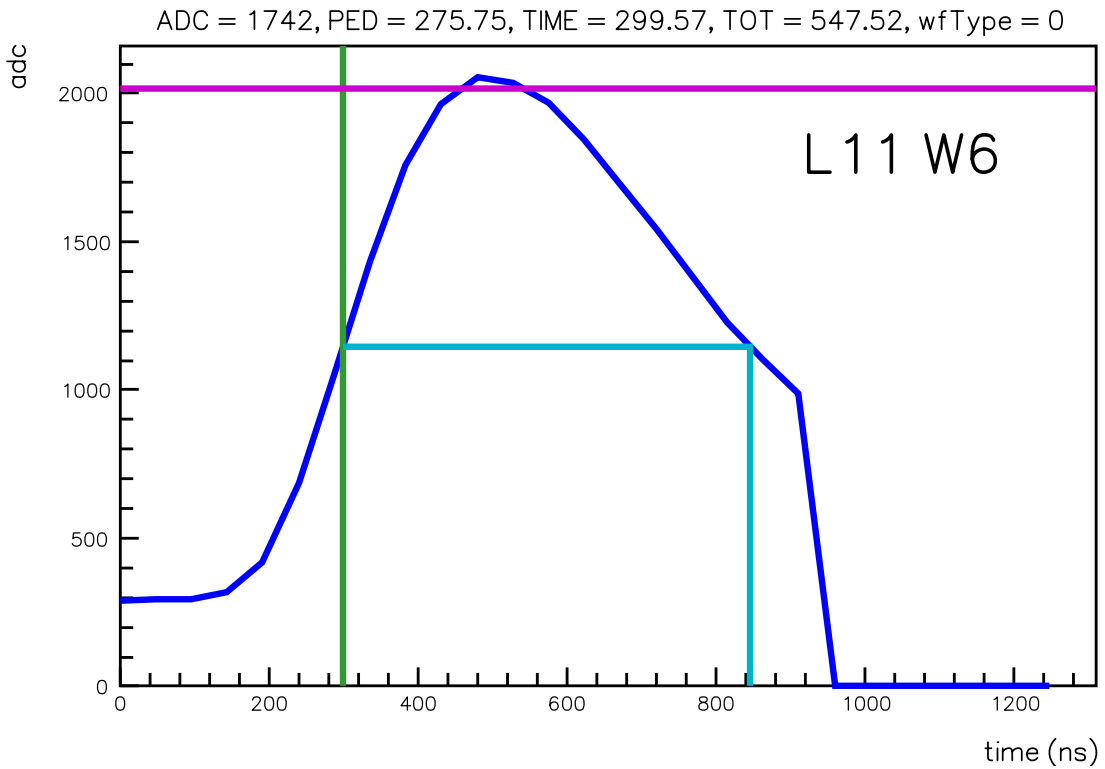




- Only signals with a good leading edge time are relevant (wfType ≤ 2)

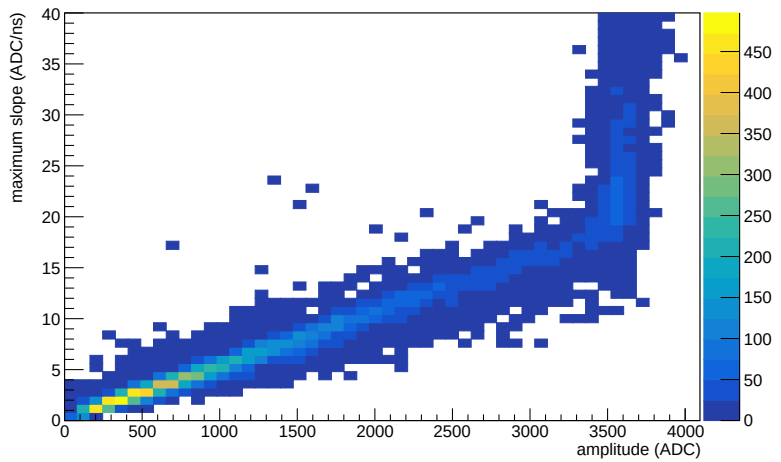
Pulse decoding

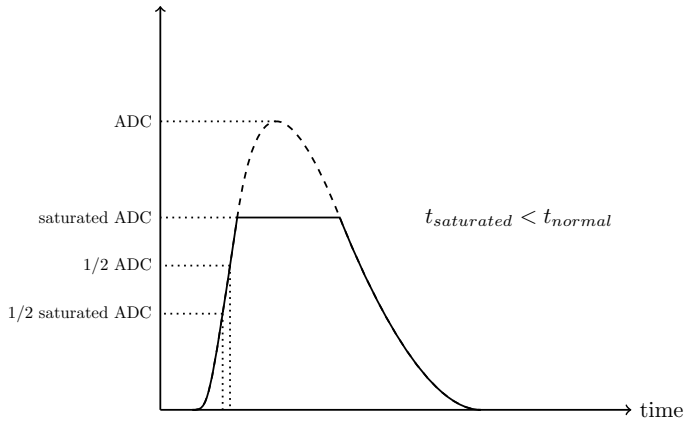
- Procedure for extracting the time
  - Baseline → estimated using the 4 first samples
  - Amplitude → after baseline subtraction, average of 3 samples around the peak
  - Leading edge time → computed at 1/2 amplitude
  - Time over threshold → computed if the signal has a trailing edge at 1/2 amplitude

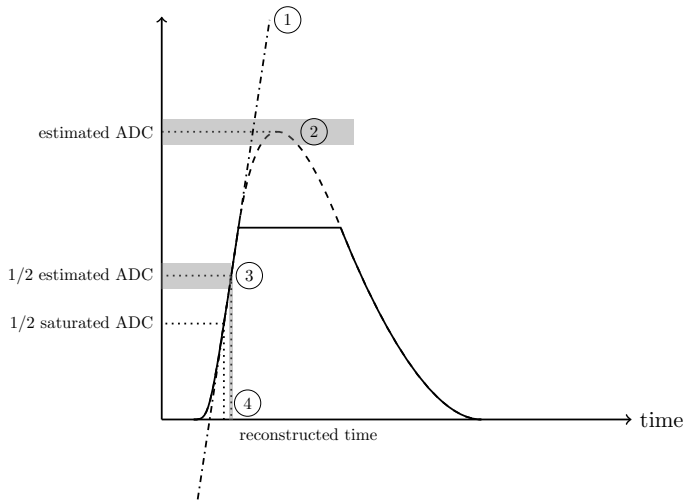


Decoding of saturated signals

- The signals of α particles always saturate
- The time extracted at 1/2 saturated ADC is under-estimated
  - It badly impacts the reconstruction
- To correct for the amplitude, we use the linear correlation of the maximum slope of a non-saturated signal with its amplitude
- With the corrected amplitude, we can recompute the time at 1/2 amplitude







- This method has been compared with the one based on the time over threshold

Typical hit selection

- Electron (pid=11, W^2 cut for elastics)
- Track (AHDC::kftrack)
  - nb. hits ≥ 7
  - $\chi^2 < 8$
  - $\Delta\phi$ cut for elastics
- Hit (AHDC::hits)
  - The track id should match the one from an existing track
  - Example of code :

```
int trackid = trackBank.getInt("trackid", i);
for (int j = 0; j < hitBank.getRows(); j++) {
    if (hitBank.getInt("trackid", j) == trackid) {
        double residual = hitBank.getDouble("residual", j);
        double time = hitBank.getDouble("time", j);
        int adcRow = hitBank.getShort("id", j) - 1;
        // moving from hit row to adc row !
        int wfType = adcBank.getShort("wfType", adcRow);
        double leadingEdgeTime = adcBank.getShort("leadingEdgeTime", adcRow);
        // etc.
    }
}
```